

CHAPTER 2

Shaping of the Earth's Surface

NCERT Exercise Solutions

The Big Questions	5 Q
Let's Map · Plate Tectonics	1 Q
Let's Explore · Plate Tectonics	3 Q
Let's Explore · Volcanic eruption (Fig. 2.6)	3 Q
Let's Explore · Erosion	1 Q
Let's Explore · Sundarbans delta	1 Q
Think About It · Glaciers	1 Q
Let's Explore · Landforms around you	1 Q
Let's Explore · Landforms and Disasters	5 Q
Questions and activities	15 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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? The Big Questions

1 What shapes the Earth's surface?

✔ ANSWER

- **Internal forces** (from inside the Earth): plate movement, earthquakes, volcanic eruptions, folding and faulting. They make mountains, valleys and ocean basins.
- **External forces** (on the surface): weathering, erosion and deposition by running water, glaciers, wind, waves and underground water. They slowly wear down and reshape landforms.

2 What is plate tectonics? What are the effects of plate movement?

✔ ANSWER

Plate tectonics is a theory given by W.J. Morgan. It says the Earth's outer layer (lithosphere) is broken into large and small pieces called tectonic plates, which move slowly over the semi-molten asthenosphere.

Effects: formation of mountains (e.g. Himalaya), valleys, ocean basins, mid-ocean ridges, volcanoes and earthquakes. It also explains the distribution of continents and oceans.

3 How are landforms formed and how are they classified?

✔ ANSWER

Landforms are formed by internal forces (plate movement, folding, faulting, volcanoes) and by weathering, erosion and deposition.

They can be classified by the agent of gradation that makes them:

- **Running water:** waterfall, meander, oxbow lake, delta
- **Waves and currents:** beach, cliff, cave, arch, stack
- **Glaciers:** U-shaped valley, cirque, arete, moraine
- **Wind:** yardang, deflation hollow, dunes
- **Underground water:** caves, stalactites, stalagmites, sinkholes

4 How are humans and other living beings connected to these landforms?

✔ ANSWER

- Plains, deltas and meanders have fertile soil for farming and dense settlements.
- Glaciers and rivers give fresh water; waterfalls give hydroelectric power.
- Beaches and deltas support fishing; fjords are used as harbours.
- Many landforms attract tourists and have cultural or religious importance.

- Landforms also influence climate, resources, settlements and cultures.

5 **How do disasters associated with different landforms impact human lives?**

✔ ANSWER

Landslides, avalanches, GLOFs and dust storms happen in particular landforms (hill slopes, snowy mountains, glacial lakes, deserts).

They cause loss of lives and livestock, damage houses, roads, bridges and hydel projects, cut off villages and destroy farms and livelihoods.

⭐ Let's Map · Plate Tectonics

1 **Pick any two plates from the map (Fig. 2.3) and complete the table: Name of the plate | Continents | Ocean**

✔ ANSWER

Name of the plate	Continents	Ocean
African Plate	Africa	Eastern part of the Atlantic Ocean and western part of the Indian Ocean

Indo-Australian Plate	Australia and the Indian subcontinent (India)	Indian Ocean and south-western part of the Pacific Ocean

You may pick any other two, e.g. **South American Plate:** South America; western part of the South Atlantic Ocean. **North American Plate:** North America and Greenland; western part of the North Atlantic and the Arctic Ocean.

⭐ Let's Explore · Plate Tectonics

1 **Examine the plate map (Fig. 2.3) with the earthquake and volcano map (Fig. 2.4). What correlation do you observe?**

✔ ANSWER

Most earthquakes and active volcanoes lie **along plate boundaries**, where plates meet.

They are especially crowded around the Pacific Ocean, the area called the **Ring of Fire**.

So plate movement and the distribution of earthquakes and volcanoes are closely linked.

2 **Observe the map (Fig. 2.4). Can you identify which continents and countries are located around the Ring of Fire with the help of an atlas or a globe?**

💡 SAMPLE ANSWER

Continents: Asia, North America, South America and Australia/Oceania (and Antarctica in the south).

Countries: Japan, Philippines, Indonesia, Papua New Guinea, New Zealand, Russia (Kamchatka), USA (Alaska, west coast), Canada, Mexico, Central American countries, Colombia, Ecuador, Peru and Chile.

3 **Does India have a risk of earthquakes? Can you find out which region is more vulnerable to earthquakes? Why do you think human lives are at risk?**

💡 SAMPLE ANSWER

Yes. The **Himalayan region and the North-East** are most vulnerable, because the Indo-Australian Plate is pushing into the Eurasian Plate there. Kutch (Gujarat) and the Andaman and Nicobar Islands are also high-risk (e.g. Gujarat earthquake, 2001).

Lives are at risk because India is densely populated, many buildings are not earthquake-resistant, and collapsing houses, landslides and fires kill people.

⭐ Let's Explore · Volcanic eruption (Fig. 2.6)

1 **What do you think caused this situation?**

✔ ANSWER

A **volcanic eruption** nearby. The ash thrown out covered the houses, trees and the ground.

2 **What could that grey powder be?**

✔ ANSWER

It is **volcanic ash**: tiny pieces of rock and glass thrown into the air during the eruption, which then settled down.

3 **What does it tell us about the Earth's internal forces?**

✔ ANSWER

The Earth's interior is very hot and full of energy. Molten material (magma) and gases can force their way up and burst out.

Internal forces are powerful enough to change the land and affect human life suddenly.

⭐ Let's Explore · Erosion

1 **Observe the photographs (Fig. 2.9) and note the types of erosion. How are farmers affected by erosion due to water and wind?**

✔ ANSWER

Types: (a) **water erosion** and (b) **wind erosion**.

- Water erosion (rain, rivers) washes away fertile topsoil and makes gullies, so fields become uneven and less fertile.
- Wind erosion blows away loose dry topsoil and can cover fields with sand.

In both cases crop yields fall and farmers lose income.

⭐ Let's Explore · Sundarbans delta

1 **Have you heard about the Sundarbans delta? Try and explore its uniqueness and find out why it is popular with tourists.**

💡 SAMPLE ANSWER

- It is formed by the Ganga, Brahmaputra and Meghna rivers in West Bengal (India) and Bangladesh; it is one of the largest deltas in the world.
- It has the largest mangrove forest in the world.
- It is home to the Royal Bengal Tiger, crocodiles and many birds; it is a UNESCO World Heritage Site.

Tourists come for boat safaris through the creeks, wildlife watching and the unique mangrove landscape.

🤔 Think About It · Glaciers

1 **A devastating flood struck the Chamoli district in Uttarakhand in February 2021. Can you find out the reasons that led to the sudden and unexpected flood?**

💡 SAMPLE ANSWER

- A huge mass of **rock and glacier ice broke off** a high mountain slope and fell into the valley (a rock and ice avalanche).
- It mixed with snow, water and debris and rushed down the Rishiganga and Dhauliganga rivers as a sudden flood.
- Rising temperatures (melting, weakened glaciers), steep slopes and construction in the fragile valley added to the damage.

⭐ Let's Explore · Landforms around you

1 **Observe the landforms around your school or residence and try to identify which agent may have created them.**

💡 SAMPLE ANSWER

Example (plains area): flat floodplain and a winding river near the town were made by **running water** (deposition); a small gully in a field was made by rain-water erosion.

Example (hills): V-shaped valleys and waterfalls by running water; in a desert, sand dunes by wind.

⭐ Let's Explore · Landforms and Disasters

1 **Make a list of disaster-prone areas from India and the world and enlist mitigation measures quoting recent examples.**

💡 SAMPLE ANSWER

See the four disaster boxes below (landslides, avalanches, GLOFs, dust storms). Use newspapers and atlases to add the latest examples.

Landslides **Landslides: Prone areas, Mitigation measures, Recent examples**

💡 SAMPLE ANSWER

Prone areas: Himalaya (Uttarakhand, Himachal Pradesh, Jammu and Kashmir, Sikkim), North-East, Western Ghats (Kerala, Karnataka, Maharashtra); world: Nepal, Japan, Andes, Alps.

Mitigation: planting trees, retaining walls, proper drainage, no construction on steep or loose slopes, careful road building, early warning.

Recent example: Wayanad, Kerala (July 2024), after very heavy rain.

Avalanches

Avalanches: Prone areas, Mitigation measures, Recent examples

💡 SAMPLE ANSWER

Prone areas: snowy parts of the Himalaya: Jammu and Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, Sikkim; world: Alps, Rockies.

Mitigation: snow fences and barriers, tree plantation on slopes, avalanche warnings, avoiding trekking/skiing after heavy snowfall, controlled release of snow.

Recent example: Mana, Chamoli district, Uttarakhand (February 2025), which hit a road workers' camp.

GLOFs

GLOFs: Prone areas, Mitigation measures, Recent examples

💡 SAMPLE ANSWER

Prone areas: glacial lakes of the Himalaya: Sikkim, Uttarakhand, Himachal Pradesh, Ladakh; world: Nepal, Bhutan, Andes.

Mitigation: mapping and monitoring glacial lakes, lowering lake water (siphoning), early warning systems, no settlements close to rivers below such lakes.

Recent example: South Lhonak Lake, Sikkim (October 2023), which damaged a dam on the Teesta river.

Dust Storms

Dust Storms: Prone areas, Mitigation measures, Recent examples

💡 SAMPLE ANSWER

Prone areas: Rajasthan (Thar Desert), Punjab, Haryana, Delhi, western Uttar Pradesh; world: Sahara, Middle East, Gobi.

Mitigation: shelter belts of trees, stopping overgrazing, keeping soil covered, staying indoors and using masks during storms, weather warnings.

Recent example: May 2018 dust storms in Rajasthan and Uttar Pradesh that killed many people.

? Questions and activities

1 **What are the sources of energy that are required to cause movements associated with the internal forces of the Earth?**

✔ ANSWER

- **Heat from the Earth's core** (the interior is extremely hot).
- This heat makes molten material in the mantle rise and cooler material sink, forming **convection currents**.

- These currents push and pull the tectonic plates, causing earthquakes, volcanoes, folding and faulting.

2 **Relate various physiographic divisions you have studied in the earlier grades with various endogenic forces responsible for their origin.**

💡 SAMPLE ANSWER

- **Himalayan mountains:** fold mountains formed when the Indian (Indo-Australian) Plate collided with the Eurasian Plate (convergent boundary).
- **Northern Plains:** formed in the depression (trough) between the Himalaya and the Peninsula, filled by river deposits.
- **Peninsular Plateau:** old, stable block of the land; its rift valleys (Narmada, Tapi) formed by faulting; the Deccan Traps by volcanic lava flows.
- **Islands:** Andaman and Nicobar are parts of submerged mountains near a plate boundary (with Barren Island volcano); Lakshadweep are coral islands.

3 **Why and where do earthquakes occur frequently? Is it possible to predict earthquakes?**

✔ ANSWER

Why: plates move, collide, sink or slide past each other; stress builds up and is released suddenly as an earthquake.

Where: along plate boundaries, especially the Ring of Fire around the Pacific Ocean; in India, the Himalaya and the North-East.

Prediction: we cannot predict the exact time and place of an earthquake. We can only identify earthquake-prone regions and prepare for them.

4 **"Plate movements are responsible for the distribution of earthquakes and volcanoes." Explain.**

✔ ANSWER

- Most earthquakes and volcanoes occur **along plate boundaries**, e.g. the Ring of Fire.
- **Convergent:** an oceanic plate sinks beneath a continental plate, causing volcanoes and earthquakes.
- **Divergent:** plates move apart, magma rises and forms new crust (Mid-Atlantic Ridge), with volcanic activity.
- **Transform:** plates slide past each other, causing earthquakes (San Andreas Fault, USA).

So the map of earthquakes and volcanoes closely follows the map of plate boundaries.

5 **Draw and label a diagram of a meander and a delta.**

💡 SAMPLE ANSWER

Meander (Fig. 2.11): draw a river with a big loop. Label: River, Steep bank (outer bend, erosion), Bar (inner bend, deposition), Oxbow lake (a cut-off loop).

Delta (Fig. 2.12): draw a river splitting into many channels in a triangle/fan shape as it meets the sea. Label: River, Distributary, Islands/Bars, Sea.

(check the pictures in your book)

6 **How are deforestation and erosion associated with each other? Explain.**

✔ ANSWER

- Tree roots hold the soil together and plant cover protects it from rain and wind.
- When trees are cut, the soil is left bare, so rain-water and wind easily carry the topsoil away.
- This causes more soil erosion and loss of fertility, and on slopes it can lead to landslides; in dry areas it leads to dust storms.

7 **Develop a plan to protect the land in your local area from erosion.**

💡 **SAMPLE ANSWER**

- Plant trees, grass and shrubs, especially on slopes and river banks.
- On hills, use terracing, contour trenches and bunding.
- Build small check dams across streams to slow water.
- Stop overgrazing and keep fields covered (mulch, cover crops).
- Make proper drainage so rain-water does not run fast over bare land.

8 **Which disasters do you think you might experience in your region? Discuss a mitigation plan in your classroom.**

💡 **SAMPLE ANSWER**

Example (a hilly region): landslides and floods in the monsoon.

Plan: plant trees on slopes, keep drains clean, avoid building on steep slopes, follow weather warnings, keep an emergency kit, know safe places and helpline numbers, practise drills at school.

9 **Prepare a model of landforms created by underground water.**

💡 **SAMPLE ANSWER**

Use a shoebox, clay/plaster and paint to make a limestone area:

- a **cave** with a cave mouth,
- **stalactites** hanging from the roof and **stalagmites** rising from the floor (a **pillar** where they join),
- a **sinkhole** on the surface and an **underground river** flowing through the cave.

Label each part.

10 **What precautionary measures will you take if you are staying in an earthquake-prone region?**

💡 **SAMPLE ANSWER**

- Live in earthquake-resistant buildings; fix heavy furniture to walls.
- Keep an emergency kit (water, torch, first aid, medicines).
- During shaking: **Drop, Cover and Hold** under a strong table; stay away from windows.
- If outside, move to an open space away from buildings, trees and power lines.
- After shaking: do not use lifts; follow official instructions.

11 **Prepare a map showing landform-associated disasters that happened in the current calendar year.**

💡 **SAMPLE ANSWER**

Take an outline map of India/the world. Collect news of landslides, avalanches, GLOFs, floods and dust storms this year, mark each place with a symbol (e.g. triangle for landslide), add the date, and give a key/legend.

12 **Create a poster showing landforms that are considered to be sacred or important in your region, and add the folk stories associated with them.**

💡 **SAMPLE ANSWER**

Example: rivers like the Ganga, hills like Govardhan, caves like Amarnath. Draw or paste pictures, write the name and location of each, and a short folk story told about it by elders in your area.

13 **Document a case of a disaster that hit your region in the past, highlighting its effects on various human activities.**

💡 **SAMPLE ANSWER**

Example: Gujarat earthquake, 2001 (Fig. 2.5): many deaths, houses and schools destroyed, roads and water supply damaged, shops and factories closed, farming and trade disrupted for months.

Write: what happened, when, where, effects on homes, farming, jobs, schools, transport, and how people recovered.

14 **Translate the given poster on landslide into your native language and display it in your home.**

💡 **SAMPLE ANSWER**

Activity: translate each line of the landslide poster into your language (e.g. Hindi: भूस्खलन), keep the pictures, and put it where your family can see it.

15 **Divide the class into three groups (water, wind, glacier). Each project should highlight the causes, impact on human life and the environment, and mitigation measures.**

💡 **SAMPLE ANSWER**

Group	Causes	Impact	Mitigation

Water	Heavy rain, rivers, waves erode soil	Loss of topsoil, floods, houses and roads washed away	Trees, terracing, bunds, check dams
Wind	Dry loose soil, little vegetation, strong winds	Dust storms, loss of soil, sand covering fields	Shelter belts, stop overgrazing, cover soil
Glacier	Moving ice, melting due to rising temperatures	GLOFs, avalanches, floods downstream	Monitor glacial lakes, early warning, safe settlement

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**